

Active Minimization and Typological Constraints in Syntactic Dependencies Across 12 Languages

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1 Motivation for the Research Problem

Dependency grammar maps out syntactic structures by connecting words to their heads. The linear distance between these pairs, often called dependency distance, serves as a clear indicator of cognitive load: when syntactic elements are separated by many words, working memory demands during real-time processing increase significantly. Because of this, the principle of Dependency Length Minimization (DLM) suggests that human languages naturally evolve structures that keep these connections as short as possible.

While standard metrics show that natural languages favor short dependencies (see Appendix A for global baselines), this project looks into the specific mechanisms governing these structures across 12 different languages. We aim to prove that minimization is an active process by comparing actual syntactic networks against randomized baselines and seeing how the number of intervening words scales as sentences get longer. We also analyze how different word-order types, such as SVO, SOV, or flexible systems, influence normalized distances, branching directions, and the types of words that tend to sit between dependencies.

2 Hypotheses and Predictions

Based on theories of cognitive processing and linguistic typology, we propose two primary hypotheses:

- **Hypothesis 1 (Active Minimization):** Syntactic dependencies are actively compressed by grammatical rules to reduce cognitive load. We predict that observed dependency distances will be drastically shorter than randomized baselines across all languages. Furthermore, we predict the number of intervening words will scale sublinearly with sentence length, resisting proportional growth.
- **Hypothesis 2 (Typological Constraints):** Word-order rules systematically govern how dependencies are resolved. We predict that distinct typologies (SOV, SVO, and Mixed/Flexible) will rely on different intervening part-of-speech categories and exhibit differing normalized dependency distances. Additionally, we predict these grammatical constraints will dictate dependency directionality, with verb-final (SOV) languages displaying strong rightward branching asymmetries compared to verb-medial (SVO) languages.

3 Methods

3.1 Data Collection

This study utilizes syntactically annotated treebanks from the Universal Dependencies (UD) dataset. To ensure a robust cross-linguistic sample, 12 languages were selected and grouped by dominant word order typology: SVO (English, French, Spanish, Chinese), SOV (Hindi, Japanese, Turkish, Korean), and Mixed/Flexible (German, Dutch, Finnish, Portuguese). The final parsed dataset comprises over 2.6 million valid dependency relations.

3.2 Metrics and Implementation

CoNLL-U data were processed using a custom Python pipeline (see Appendix B). For each valid relation (excluding root nodes), we extracted Universal Part-of-Speech (UPOS) tags, dependency direction (left/right), and absolute dependency distance ($D = |H_{\text{idx}} - D_{\text{idx}}|$). To enable fair cross-linguistic comparison despite varying natural sentence lengths, we also calculated a *Normalized Distance* by dividing the absolute distance by the total sentence length.

3.3 Inference and Baseline Generation

To test for active distance minimization, we generated a randomized baseline for every sentence. Word order was algorithmically shuffled while perfectly preserving the original sentence length, creating a null distribution of syntactic distances governed by chance. The extracted data were then aggregated globally to assess baseline scaling effects, and subsequently partitioned by typology (SVO, SOV, Mixed) to evaluate systematic variations in distance, part-of-speech reliance, and structural directionality.

4 Results

4.1 Baseline Global Constraints: Distance and POS Distributions

To establish a universal structural baseline, we aggregated over 2.6 million dependency relations across all 12 languages. The analysis robustly supports the presence of universal Dependency Length Minimization (DLM). Globally, the median number of intervening words is a mere 1.00, with 75% of all dependencies featuring two or fewer intervening elements (see Appendix Figure 8).

While the global mean absolute dependency distance is tightly constrained at 3.33, individual languages exhibit a distinct distributional range (see Appendix Figure 9). Korean (2.43) and Japanese (2.82) maintain the tightest absolute networks, whereas Chinese (3.98) and German (3.77) display the longest average distances before length normalization.

Furthermore, the structural space between these dependencies is not occupied randomly. As illustrated in Appendix Figure 7, the distribution is heavily skewed toward core lexical and structural anchors. Nouns (NOUN) and Adpositions (ADP) are the most frequent intervening elements, followed closely by functional categories like Determiners (DET) and Proper Nouns.

4.2 Objective 1: Systematic Typological Differences

Disaggregating the data by word-order typology (SVO, SOV, and Mixed/Flexible) revealed significant, systematic differences in both dependency distance and the types of

intervening words. Statistically, Subject-Object-Verb (SOV) languages exhibited tighter absolute syntactic structures, with a shorter mean dependency distance of 2.94, compared to Subject-Verb-Object (SVO) languages, which averaged a distance of 3.52. However, when assessing *Normalized Distance* (which controls for varying natural sentence lengths), SVO languages actually formed the tightest structures (0.12), followed by SOV (0.15). Crucially, the Mixed/Flexible typology exhibited the loosest structural networks overall (0.17).

Furthermore, Figure 1 demonstrates the profound impact of typology on the distribution of intervening POS categories. While Nouns (NOUN) consistently account for a large portion of interveners across all language families, their concentration is particularly extreme in SOV languages like Korean (36.3%) and Turkish (31.6%). The heatmap also reveals distinct functional shifts: Hindi and Japanese (both SOV) show a heavy reliance on Adpositions (28.1% and 29.1%, respectively). In sharp contrast, SVO languages—particularly French and Spanish—display a stark structural shift toward Determiners (DET), which account for over 20% of their intervening words, compared to less than 2.5% in most SOV languages.

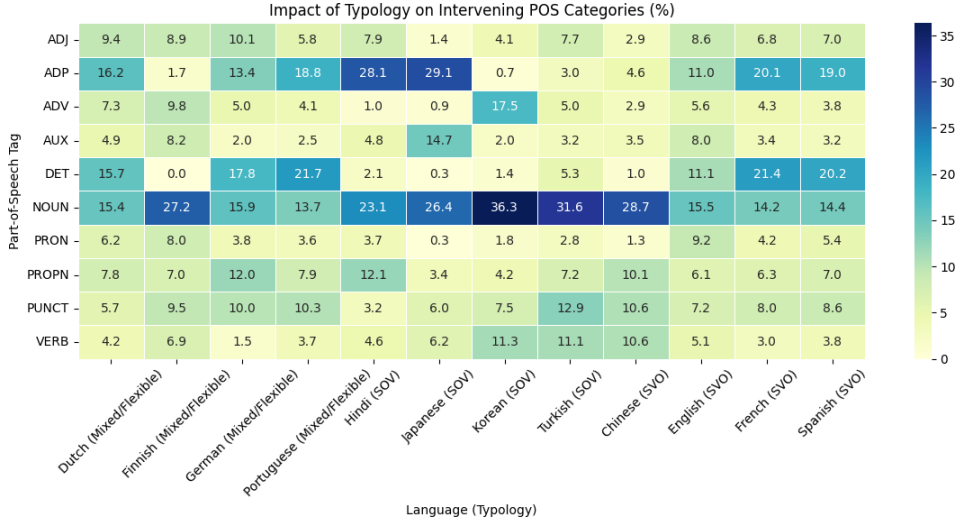


Figure 1: Distribution of Intervening POS Categories by Typology (% of dependencies)

4.3 Objective 2: Assess Global Dependency Directionality (Left vs. Right Branching)

To further understand syntactic structure, we analyzed whether dependents occur to the left or right of their syntactic heads. Across the entire 12-language dataset, dependency direction exhibits a clear asymmetry: rightward dependencies dominate globally, accounting for 60.45% of all relations, compared to 39.55% for leftward dependencies.

As shown in Figure 2, this pattern holds across most individual languages. Languages such as Korean (75.05%), Turkish (68.57%), and German (63.83%) exhibit particularly strong right-branching tendencies. In contrast, Japanese presents a more balanced distribution, with a slight preference for leftward dependencies (51.72%).

These findings suggest that while a universal preference for rightward dependencies exists, languages retain flexibility depending on their specific grammatical structures and typological constraints.

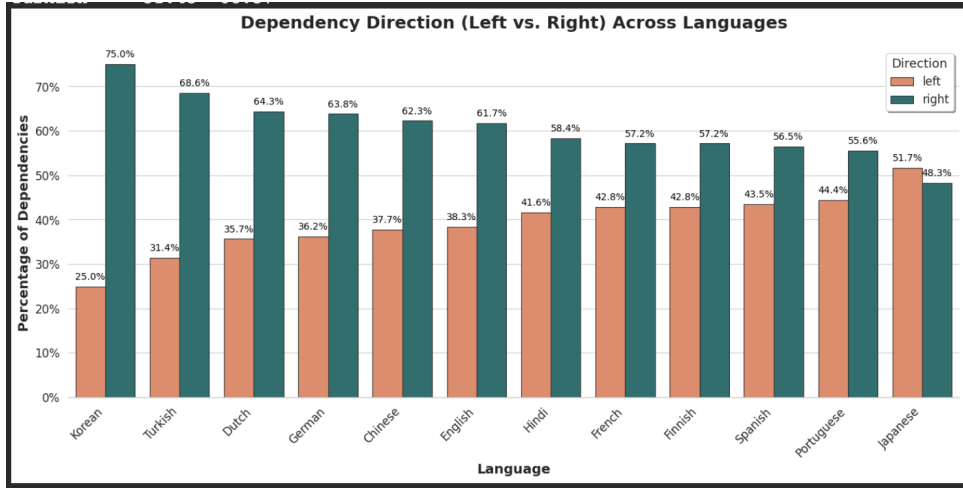


Figure 2: Dependency Direction (Left vs. Right) Across 12 Languages

4.4 Objective 3: Assess Dependency Directionality Grouped by Typology

To examine whether dependency direction is influenced by grammatical structure, we grouped the dataset into SVO, SOV, and Mixed/Flexible typologies. The results reveal systematic typological variation in directional preferences.

As illustrated in Figure 3, SOV languages exhibit the strongest rightward bias, with 63.25% of dependencies branching to the right. Mixed/Flexible languages show a similar pattern (60.75%). In contrast, SVO languages demonstrate a comparatively weaker rightward preference (58.16%) and the highest proportion of leftward dependencies (41.84%).

This pattern aligns strongly with syntactic expectations: verb-final (SOV) languages naturally accumulate dependents before the verb, resulting in a higher concentration of rightward dependencies relative to earlier elements. SVO languages, by virtue of their verb-medial structure, distribute syntactic relations more evenly across both directions.

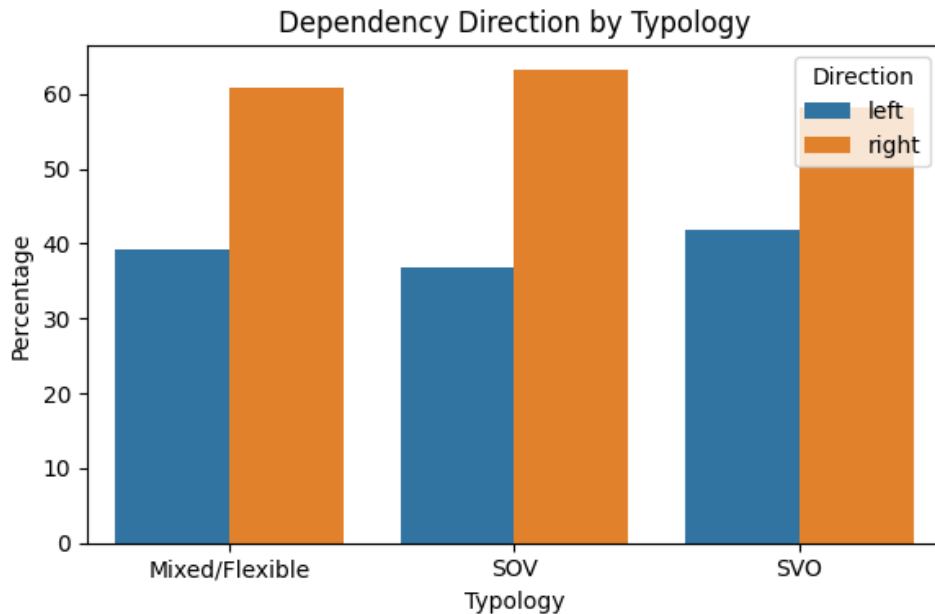


Figure 3: Dependency Direction (Left vs. Right) Grouped by Typology

4.5 Objective 4: Evidence of Active Length Minimization

To directly test whether Dependency Length Minimization is an actively enforced structural constraint, we compared observed dependency distances with a randomized baseline. In the randomized condition, word order was shuffled while preserving sentence length, generating a null distribution.

The results reveal a marked contrast: the global mean real dependency distance is 3.33, whereas the randomized distance averages nearly 9.90. This nearly threefold reduction in distance demonstrates that natural languages strongly avoid long dependencies compared to chance (Figure 4).

To ensure this global effect is not a statistical artifact, we assessed the real versus random baseline at the individual language level. As illustrated in Figure 5, the pattern is universally consistent: in every language analyzed, real dependency distances are significantly shorter than their randomized counterparts. For example, Chinese shows a real mean distance of 3.98 compared to 10.69 under randomization, while Japanese exhibits 2.82 versus 11.29. Notably, languages with longer real dependencies also exhibit proportionally larger random baselines, reinforcing the strict universality of the DLM constraint across distinct grammars.

Furthermore, evaluating how intervening words scale with sentence length proves the resilience of this minimization. While the absolute number of intervening words inevitably increases as sentences grow, the scaling is heavily suppressed. For short sentences (1–10 words), the mean number of intervening words is 1.05. Even as sentences expand to 40+ words, the mean number of intervening words only scales to 3.08. As demonstrated in Figure 6, this sublinear scaling proves that language processing mechanisms aggressively cluster dependencies regardless of overall utterance complexity.

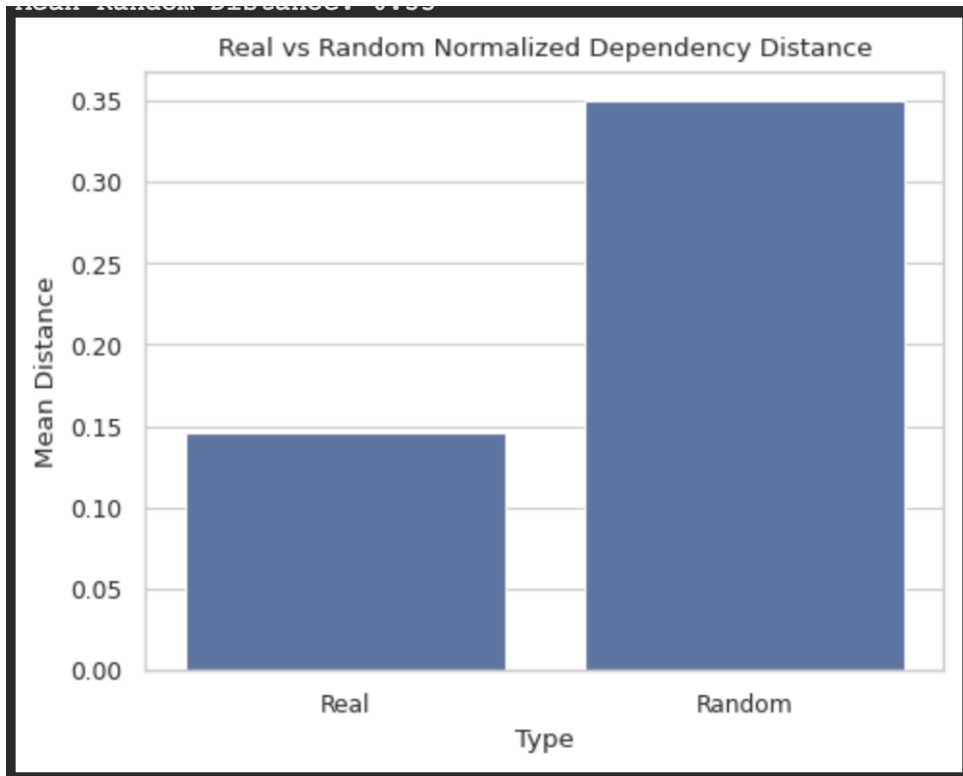


Figure 4: Global Real vs. Random Dependency Distance

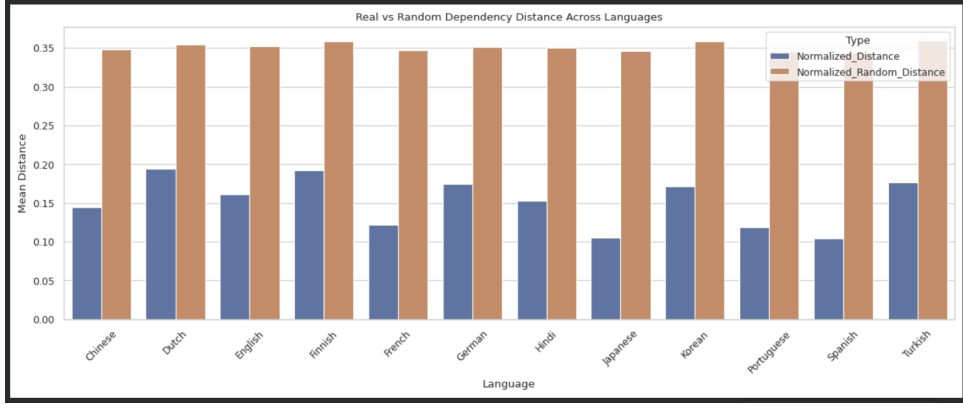


Figure 5: Real vs. Random Dependency Distance Across 12 Languages

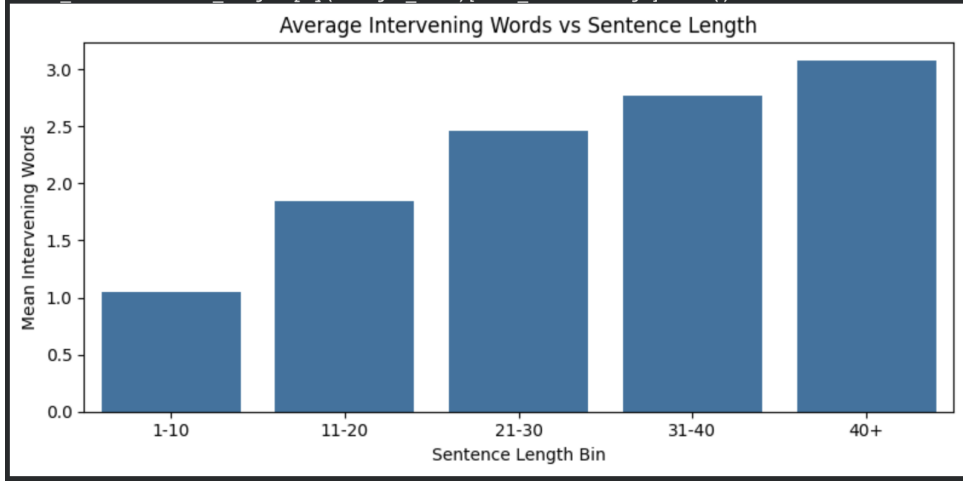


Figure 6: Average Intervening Words Scaling Sublinearly with Sentence Length

5 Theoretical Implications

These findings empirically validate active Dependency Length Minimization (DLM), confirming that grammatical rules actively compress syntactic structures to mitigate working memory constraints during real-time comprehension. Furthermore, the sublinear scaling of intervening words in longer sentences demonstrates a universal cognitive ceiling on holding unattached linguistic elements in active memory.

Crucially, languages navigate these universal cognitive limits through distinct evolutionary strategies. Verb-final (SOV) languages pack dependencies densely with a strong rightward bias, while verb-medial (SVO) languages distribute the structural load more evenly across directions using determiners as buffers. Most notably, our analysis reveals a fascinating evolutionary trade-off: increased word order flexibility inherently comes at the cost of looser dependency networks and marginally higher baseline processing demands.

References

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A Baseline Global Descriptive Statistics

A.1 Frequent Intervening POS Categories

To determine which part-of-speech categories most frequently appear as intervening words between a syntactic head and its dependent, we aggregated the data across all 12 languages (comprising over 2.6 million total instances). As illustrated in Figure 7, the distribution is heavily skewed toward a few dominant categories. Nouns (NOUN) and Adpositions (ADP) are the most frequent intervening elements by a significant margin. They are followed closely by functional categories like Determiners (DET) and Proper Nouns (PROPN). This robustly supports our hypothesis that core lexical and functional categories predominantly occupy the structural space between syntactic dependencies.

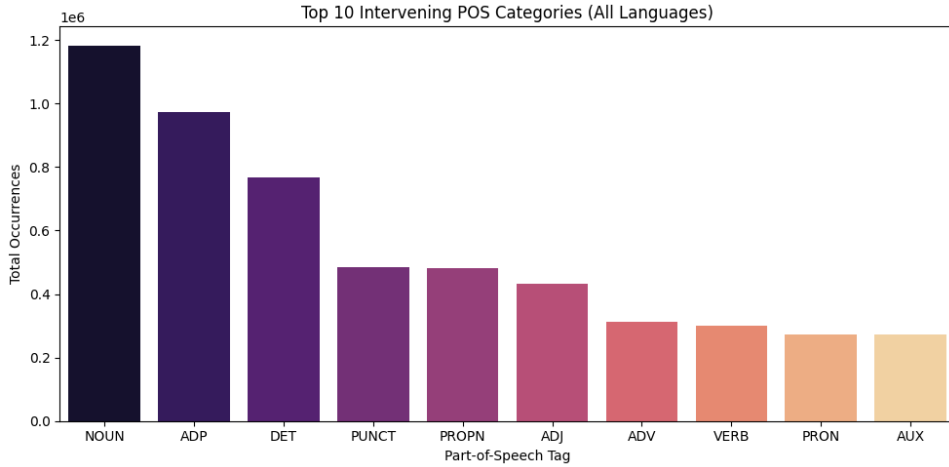


Figure 7: Universal Top 10 Intervening POS Categories (All 12 Languages Combined)

A.2 Typical Number of Intervening Words

The aggregated analysis strongly supports the principle of Dependency Length Minimization (DLM). Across the entire dataset of 2,600,038 valid dependency relations, the global average number of intervening words occurring between a syntactic head and its dependent is exceptionally low at 2.33 ($SD = 5.01$).

As illustrated in Figure 8, the median number of intervening words is a mere 1.00, and 75% of all dependencies across the 12 languages feature 2 or fewer intervening words. While extreme outliers exist in the dataset, the overwhelming concentration of data at the lower bound visually demonstrates a universal typological preference for keeping syntactic dependencies as short as possible to reduce cognitive processing load.

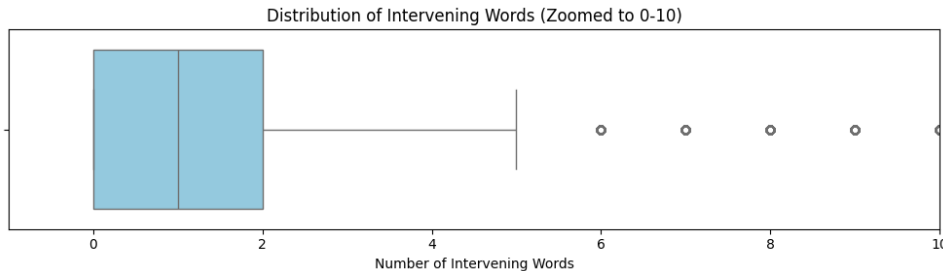


Figure 8: Distribution of Intervening Words (Zoomed to 0–10 Words)

A.3 Distribution of Dependency Distances

While the number of intervening words remained universally low, the absolute dependency distance (which accounts for the specific index positions of the head and dependent) varied significantly across the dataset. The global mean dependency distance across all languages was calculated at 3.33. However, breaking this down by individual languages revealed a distinct distributional range.

As illustrated in Figure 9, Korean (2.43) and Japanese (2.82) exhibited the tightest dependency networks, keeping structural elements significantly closer than the global average. Conversely, Chinese (3.98) and German (3.77) displayed the longest average absolute dependency distances in the sample before length normalization. These distinct distributional shifts confirm that while Dependency Length Minimization is a universal pressure, language-specific grammatical rules exert a measurable counter-influence on dependency distances.

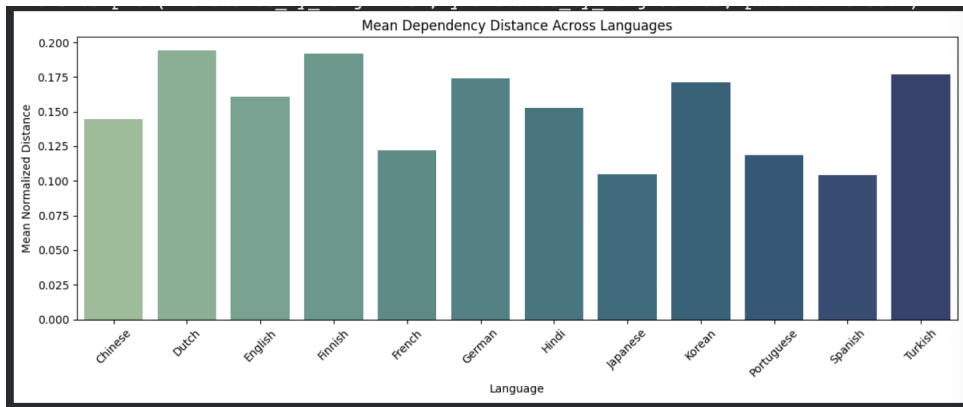


Figure 9: Mean Normalized Dependency Distance Across Languages

B Implementation Code

The complete Python implementation for this study—including data collection from the Universal Dependencies (UD) treebanks, CoNLL-U parsing, baseline randomization, and metric extraction—is hosted in a reproducible Google Colab environment.

The interactive notebook and raw scripts can be accessed at the following link: https://colab.research.google.com/drive/1_4K7tAqVfQv62o4pyMmp8TIG8tEMMHDQ?usp=sharing

C Supplementary Baselines: Real vs. Random

To further validate that observed syntactic patterns are actively enforced grammatical constraints rather than statistical artifacts, we compared real linguistic data against a randomized word-order baseline (which preserves sentence length but destroys grammatical rules).

C.1 Directionality (Left vs. Right)

In real linguistic data, typologies dictate strong structural asymmetries [2]. For example, verb-final (SOV) languages actively enforce a heavy right-branching bias, while SVO languages maintain a more balanced distribution [2]. Under randomization, however, these distinct typological signatures completely collapse. As illustrated in Figure 10, the directional distribution across all typologies normalizes, drifting toward a structural

equilibrium driven purely by chance. This definitively proves that the directional asymmetries observed in natural languages are not statistical artifacts but are actively enforced by language-specific constraints [8].

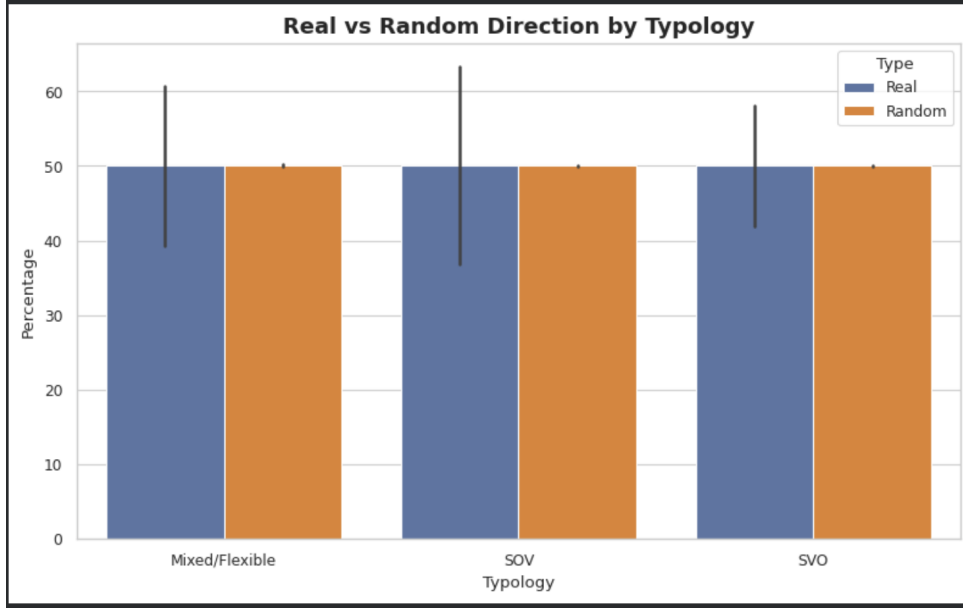


Figure 10: Real vs. Random Directional Distribution by Typology

C.2 Intervening POS Categories

Similarly, the structural space between dependencies is not populated randomly. In the real dataset, core structural anchors like Nouns (NOUN) and Adpositions (ADP) heavily dominate intervening spaces. When syntactic constraints are removed and intervening elements are sampled randomly, the distribution shifts to simply mirror the general corpus frequency of all POS categories. This divergence confirms that human language actively selects and filters the precise grammatical categories permitted to occupy the intervening space, rather than just minimizing distance.